Handwritten Digit Recognition

A Synopsis

for

Minor Project

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

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Jan-May 2026

Introduction

Handwritten Digit Recognition is a key problem in computer vision that involves identifying digits (0–9) from image data. It is widely used in applications such as cheque processing, postal services, and automated data entry systems.
Convolutional Neural Networks (CNNs) have proven to be highly effective for such tasks due to their ability to automatically extract features from images. In this project, a CNN model is developed using TensorFlow and trained on the MNIST dataset to achieve high accuracy in digit classification.
The model is integrated into a Flask-based web application, allowing users to upload images and receive real-time predictions, demonstrating both machine learning implementation and deployment.

Literature Review

Several studies have explored digit recognition using machine learning and deep learning techniques. Traditional methods such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) were initially used for classification tasks but had limitations in handling large image datasets.
Deep learning approaches, particularly CNNs, have significantly improved accuracy in image recognition tasks [1]. Research shows that CNN-based models can achieve accuracy above 97% on the MNIST dataset [2]. Advanced architectures further improve performance by optimizing layers and activation functions.
Recent developments also focus on deploying trained models into web-based systems, enabling real-time interaction and accessibility [3].

Problem Definition

Manual recognition of handwritten digits is time-consuming and prone to human error. There is a need for an automated system that can accurately and efficiently recognize handwritten digits from images.
The problem addressed in this project is to design and implement a system that:

- Accurately classifies handwritten digits (0–9).
- Processes image input from users.
- Provides real-time predictions through a web interface

Objectives

- To understand and implement Convolutional Neural Networks for image classification.
- To train a model using the MNIST handwritten digit dataset.
- To achieve high accuracy in digit recognition.
- To develop a web-based interface using Flask for user interaction.
- To integrate the trained model with the frontend for real-time predictions.
- To improve model performance by expanding and optimizing the training dataset

Methodology

The project follows these steps:

1. Data Collection

- The MNIST dataset is used, consisting of 60,000 training and 10,000 testing images.

2. Data Preprocessing

- Images are converted to grayscale
- Resized to 28×28 pixels
- Normalized to values between 0 and 1

3. Model Development

- A CNN model is designed with:
  - Convolution layers
  - Activation functions (ReLU)
  - Pooling layers
  - Fully connected dense layers
- Softmax activation is used in the output layer for classification.

4. Model Training

- Model is trained using:
  - Optimizer: Adam
  - Loss Function: Categorical Crossentropy
- Achieves high accuracy (~97–98%)

5. Model Deployment

- The trained model is saved as a .h5 file
- Integrated into a Flask backend

6. Web Application

- Users upload an image via a web interface
- Image is preprocessed and passed to the model
- Predicted digit is displayed on the webpage

7. Enhancement

- Ongoing work includes dataset expansion and performance optimization to improve generalization.

References

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- [3] Goodfellow I., Bengio Y., Courville A., Deep Learning, MIT Press, pp. 326–366, 2016.
- TensorFlow Documentation: https://www.tensorflow.org/api_docs
- Flask Documentation: https://flask.palletsprojects.com/
- MNIST Dataset: http://yann.lecun.com/exdb/mnist/

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